OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

10.1 Introduction to roads

Elements in UML model

`<road>` element

In ASAM OpenDRIVE, roads are represented by `<road>` elements within the `<OpenDRIVE>` element.

```
UML class: t_road
XML tag:   <road> (Multiplicity: 1..*)
```

Roads are the core elements for any road network in ASAM OpenDRIVE. Each road runs along one road reference line.

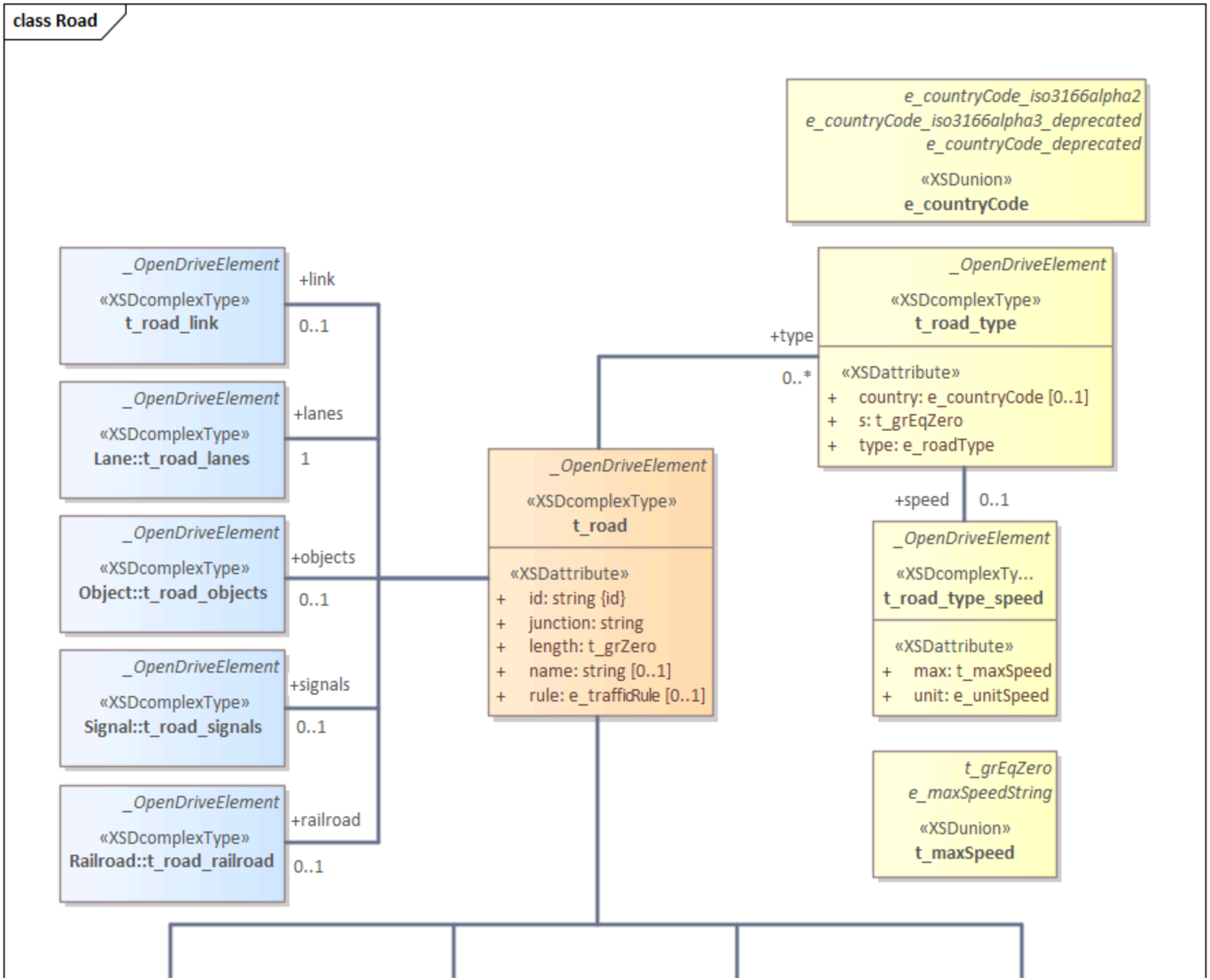
A road shall have at least the center lane. Vehicles may drive in both directions of the road reference line. The standard driving direction is defined by the value which is assigned to the @rule attribute (RHT=right-hand traffic, LHT=left-han traffic).

ASAM OpenDRIVE roads may be roads in the real road network or artificial road network created for application use. Each road is described by one or more `<road>` elements. One `<road>` element may cover a long stretch of a road, shorter stretches between junctions, or even several roads. A new `<road>` element should only start if the properties of the road cannot be described within the previous `<road>` element or if a junction is required.d

Table 23. Attributes of the <road> element

Name	Type	Use	Unit	Description
`id`	string	required		Unique ID within the database. If it represents an integer number, it should comply to uint32_t and stay within the given range.
`junction`	string	required		ID of the junction to which the road belongs, for example connecting roads, cross paths, and roads of a junction boundary. Use -1 for none.

Name	Type	Use	Unit	Description
`length`	t_grZero	required	m	Total length of the reference line in the xy-plane. Change in length due to elevation is not considered
`name`	string	optional		Name of the road. May be chosen freely.
`rule`	e_trafficRule	optional		Basic rule for using the road; RHT=right-hand traffic, LHT=left-hand traffic. When this attribute is missing, RHT is assumed.


class Road
e_countryCode_iso3166alpha2
e_countryCode_iso3166alpha3_deprecated
e_countryCode_deprecated
«XSDunion»
e_countryCode
_OpenDriveElement
«XSDcomplexType»
t_road_link
+link
0..1
_OpenDriveElement
«XSDcomplexType»
Lane::t_road_lanes
+lanes
1
_OpenDriveElement
«XSDcomplexType»
Object::t_road_objects
+objects
0..1
_OpenDriveElement
«XSDcomplexType»
Signal::t_road_signals
+signals
0..1
_OpenDriveElement
«XSDcomplexType»
Railroad::t_road_railroad
+railroad
0..1
_OpenDriveElement
«XSDcomplexType»
t_road
«XSDattribute»
+ id: string {id}
+ junction: string
+ length: t_grZero
+ name: string [0..1]
+ rule: e_trafficRule [0..1]
+type
0..*
_OpenDriveElement
«XSDcomplexType»
t_road_type
«XSDattribute»
+ country: e_countryCode [0..1]
+ s: t_grEqZero
+ type: e_roadType
+speed
0..1
_OpenDriveElement
«XSDcomplexTy...
t_road_type_speed
«XSDattribute»
+ max: t_maxSpeed
+ unit: e_unitSpeed
t_grEqZero
e_maxSpeedString
«XSDunion»
t_maxSpeed

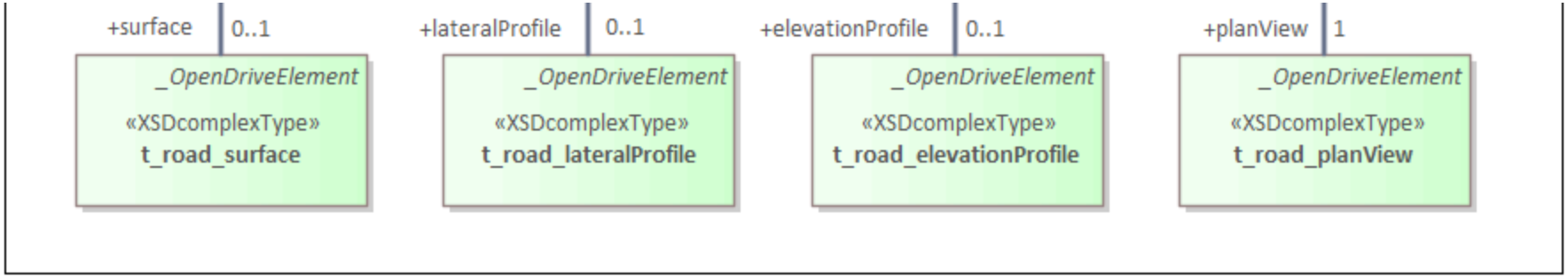
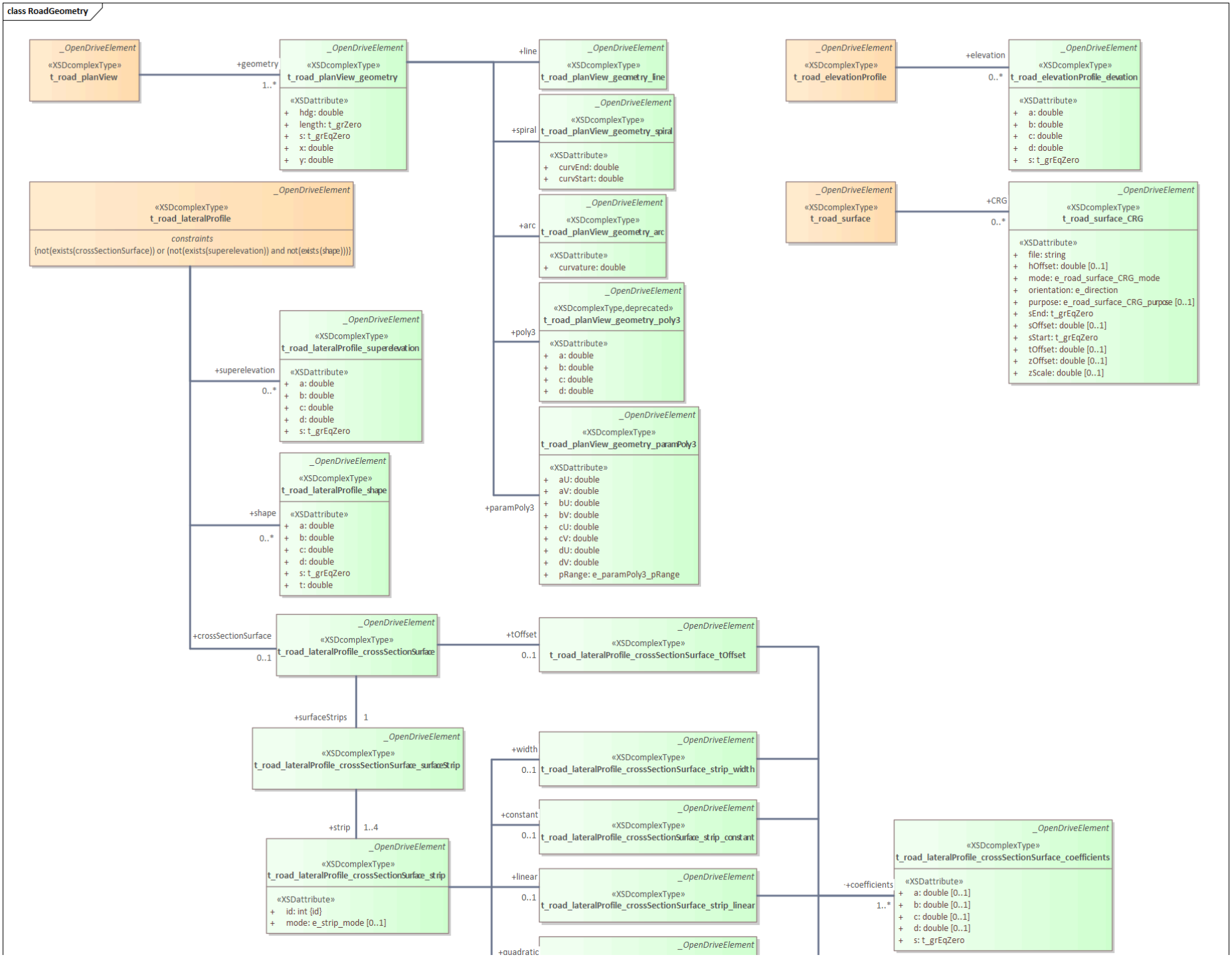


Figure 36. UML class diagram of the Road class

Figure 36 shows the UML class diagram of the ASAM OpenDRIVE Road class.


class RoadGeometry
_OpenDriveElement
«XSDcomplexType»
t_road_planView
+geometry
1..*
_OpenDriveElement
«XSDcomplexType»
t_road_planView_geometry
«XSDattribute»
+ hdg: double
+ length: t_grZero
+ s: t_grEqZero
+ x: double
+ y: double
+line
_OpenDriveElement
«XSDcomplexType»
t_road_planView_geometry_line
+spiral
_OpenDriveElement
«XSDcomplexType»
t_road_planView_geometry_spiral
«XSDattribute»
+ curvEnd: double
+ curvStart: double
+arc
_OpenDriveElement
«XSDcomplexType»
t_road_planView_geometry_arc
«XSDattribute»
+ curvature: double
+poly3
_OpenDriveElement
«XSDcomplexType,deprecated»
t_road_planView_geometry_poly3
«XSDattribute»
+ a: double
+ b: double
+ c: double
+ d: double
+paramPoly3
_OpenDriveElement
«XSDcomplexType»
t_road_planView_geometry_paramPoly3
«XSDattribute»
+ aU: double
+ aV: double
+ bU: double
+ bV: double
+ cU: double
+ cV: double
+ dU: double
+ dV: double
+ pRange: e_paramPoly3_pRange
_OpenDriveElement
«XSDcomplexType»
t_road_elevationProfile
+elevation
0..*
_OpenDriveElement
«XSDcomplexType»
t_road_elevationProfile_elevation
«XSDattribute»
+ a: double
+ b: double
+ c: double
+ d: double
+ s: t_grEqZero
_OpenDriveElement
«XSDcomplexType»
t_road_surface
+CRG
0..*
_OpenDriveElement
«XSDcomplexType»
t_road_surface_CRG
«XSDattribute»
+ file: string
+ hOffset: double [0..1]
+ mode: e_road_surface_CRG_mode
+ orientation: e_direction
+ purpose: e_road_surface_CRG_purpose [0..1]
+ sEnd: t_grEqZero
+ sOffset: double [0..1]
+ sStart: t_grEqZero
+ tOffset: double [0..1]
+ zOffset: double [0..1]
+ zScale: double [0..1]
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile
constraints
{not(exists(crossSectionSurface)) or (not(exists(superelevation)) and not(exists(shape)))}
+superelevation
0..*
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_superelevation
«XSDattribute»
+ a: double
+ b: double
+ c: double
+ d: double
+ s: t_grEqZero
+shape
0..*
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_shape
«XSDattribute»
+ a: double
+ b: double
+ c: double
+ d: double
+ s: t_grEqZero
+ t: double
+crossSectionSurface
0..1
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface
+tOffset
0..1
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_tOffset
+surfaceStrips
1
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_surfaceStrip
+strip
1..4
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_strip
«XSDattribute»
+ id: int {id}
+ mode: e_strip_mode [0..1]
+width
0..1
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_strip_width
+constant
0..1
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_strip_constant
+linear
0..1
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_strip_linear
+quadratic
_OpenDriveElement
+coefficients
1..*
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_coefficients
«XSDattribute»
+ a: double [0..1]
+ b: double [0..1]
+ c: double [0..1]
+ d: double [0..1]
+ s: t_grEqZero

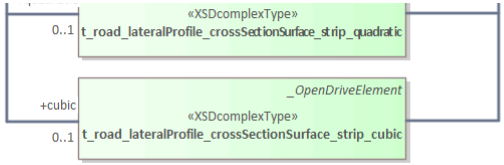


Figure 37. UML class diagram of the RoadGeometry class

Figure 37 shows the UML class diagram of the ASAM OpenDRIVE RoadGeometry class.

Rules

The following rules apply to roads:

- asam.net:xodr:1.4.0:road.overlap_inside_junction: Only roads with the same junction id may overlap on the same level. This does not include roads on different driving levels, for example, bridges.
- asam.net:xodr:1.4.0:road.no_overlap_outside_junction: Roads outside a junction shall not overlap.
- asam.net:xodr:1.4.0:road.no_overlap_self: A road shall not overlap with itself.
- The road length should be the sum of the lengths of all `<geometry>` elements

Related topics

- Section 9.2, "Road reference line"
- Section 10.3, "Road linkage"
- Section 11.1, "Introduction to lanes"